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# Distance-Based Devaluation: Do LLMs Model Persona-Persona Dislike?

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## Abstract

Large Language Models (LLMs) are increasingly used as agents in social simulations, yet their ability to model complex interpersonal dynamics remains under-explored. This paper investigates whether LLMs capture the relational structure of persona-persona dislike, hypothesizing that personas separated by greater distance in a latent persona space will express stronger antipathy toward each other’s preferences. Using the NEMOTRON-PERSONAS-USA dataset and GPT-4O-MINI, we map personas into a multidimensional embedding space and conduct reciprocal preference evaluations. Our results demonstrate a statistically significant negative correlation ( $r = -0.49$ ,  $p < 0.001$ ) between persona distance and reciprocal preference agreement, indicating that LLMs inherently encode an “us-vs-them” logic based on value and lifestyle divergence. Furthermore, we introduce zero-shot schismogenesis, a task where the model adopts the “complete opposite” of a target persona. In this setting, the model demonstrates a remarkable 88% rejection rate of the target’s preferences, justifying its opposition through structural counter-identities. These findings suggest that LLMs do not merely model isolated traits but possess a sophisticated social map where identity is defined through relational opposition, with significant implications for polarization and cooperative AI design.

## 1 Introduction

As Large Language Models (LLMs) move from simple text generators to autonomous agents in multi-agent simulations and social modeling, understanding their internal representation of social dynamics becomes critical. Humans do not exist in isolation; our identities are defined not just by what we value, but by what we oppose. This relational aspect of identity—the “us-vs-them” dynamic—is a fundamental driver of human social behavior, polarization, and group formation.

**Why does this matter?** If LLMs inherently model “dislike” based on persona distance, they might inadvertently amplify polarization or fail to find common ground in multi-agent settings. Conversely, if they capture the relational logic of opposition, they could be powerful tools for understanding and mitigating social conflict. Capturing this “relational” aspect of identity is a key step toward more sophisticated and human-like social modeling.

However, current research largely focuses on well-known social divides such as politics, race, or religion Dong et al. [2024b], Prama et al. [2025]. There is limited work on whether LLMs capture “persona-persona dislike” in a more fluid, preference-driven space. Specifically, does a distance in persona space—defined by disparate hobbies, values, and lifestyle—automatically translate into mutual devaluation of each other’s specific preferences, even if those preferences aren’t inherently controversial?

We hypothesize that LLMs do not just model isolated persona traits but also encode the relational logic of opposition. To test this, we move beyond demographic and political divides to investigate whether LLMs exhibit “relational dislike” based on latent persona distance. We use high-fidelity,

multifaceted personas from NEMOTRON-PERSONAS-USA and test if the degree of "dislike" for another persona's preferences correlates with the distance between them in a multidimensional persona space (see figure 1).

Our main finding is a strong and significant negative correlation ( $r = -0.4913$ ,  $p = 0.0003$ ) between persona distance and preference agreement. Furthermore, we demonstrate that LLMs can perform zero-shot schismogenesis with an 88% rejection rate, showing that the "opposite" relationship is structurally encoded.

Our contributions are as follows:

- We demonstrate that LLMs capture a "social map" where latent distance between personas translates into perceived opposition and preference devaluation.
- We introduce the Zero-shot schismogenesis task to evaluate whether LLMs can systematically invert a target's preferences by constructing a counter-identity.
- We conduct a quantitative analysis showing a significant correlation between embedding-based persona distance and Likert-scale preference agreement.
- We provide qualitative evidence that LLMs justify their "dislike" through nuanced value-based contrasts rather than simple sentiment shifts.

## 2 Related Work

The study of persona-conditioned LLMs has revealed their ability to adopt distinct social identities and exhibit human-like biases. We categorize related work into intergroup bias in LLMs and persona-steered generation.

**Intergroup Bias in LLMs.** Recent work has quantified "us-versus-them" dynamics in LLMs. Prama et al. [2025] demonstrate consistent ingroup favoritism and outgroup hostility across multiple architectures, finding that conservative personas often exhibit higher outgroup hostility. Similarly, Dong et al. [2024b] show that LLMs emulate "partisan sorting," where identities like race and religion align with partisan views, leading to reciprocal opposition. Dong et al. [2024a] further highlight that outgroup bias (negativity toward "them") can be even more dominant than ingroup favoritism. Our work builds on these findings by moving beyond predefined social categories to a more fluid, embedding-based persona distance.

**Persona-Steered Generation and Sentiments.** LLMs are capable of adopting complex personas that reflect real-world social sentiments. Tanaka et al. [2024] validate that LLMs encode "inter-demographic sentiments" that align with societal surveys. However, steering models toward specific personas can also amplify existing biases Liu et al. [2024]. While these studies focus on demographic groups, our research investigates whether purely preference-based distance leads to similar "dislike" dynamics.

**Schismogenesis.** Originally coined in anthropology to describe the process of differentiation in the norms of individual behavior resulting from cumulative interaction between individuals, we apply the term to LLMs. While others have explored identity adoption, we explicitly test "zero-shot schismogenesis"—the ability of a model to define its identity purely in opposition to another persona's preferences.

## 3 Methodology

We investigate persona-persona dislike through a two-pronged approach: (1) measuring the correlation between persona distance and preference devaluation, and (2) testing for zero-shot schismogenesis.

### 3.1 Persona Selection and Distance Mapping

We sample 20 multifaceted personas from the NEMOTRON-PERSONAS-USA dataset. Each persona includes a summary, cultural background, and professional bio. To quantify the "social distance" between personas, we compute embeddings for the full descriptions using the ALL-MINI-LM-L6-v2 model. We define the distance  $d(P_A, P_B)$  between Persona  $A$  and Persona  $B$  as the cosine distance

between their respective embeddings:

$$d(P_A, P_B) = 1 - \frac{\mathbf{v}_A \cdot \mathbf{v}_B}{\|\mathbf{v}_A\| \|\mathbf{v}_B\|} \quad (1)$$

where  $\mathbf{v}_A$  and  $\mathbf{v}_B$  are the embedding vectors.

### 3.2 Preference Generation and Reciprocal Evaluation

For each persona, we use GPT-4O-MINI to generate 5 specific "likes" and 5 specific "dislikes" based on their description. These preferences are intended to be grounded in their lifestyle and values.

In the evaluation phase, we select 10 personas to act as "Evaluators." Each Evaluator is prompted to adopt their assigned persona and rate their agreement with the preferences of 6 target personas on a 1–5 Likert scale. The targets are selected to cover a range of distances:

- **Self:** The Evaluator’s own preferences (baseline).
- **Close:** Two target personas with small  $d(P_A, P_B)$ .
- **Medium:** One target persona with moderate distance.
- **Far:** Two target personas with maximum distance in the sampled set.

### 3.3 Zero-shot schismogenesis Test

To test if LLMs capture the structural logic of opposition, we conduct a "zero-shot schismogenesis" test. The model is instructed: "You are the complete opposite of [Target Persona Description]. Rate your agreement with the following preferences of [Target Persona]." We measure the rejection rate (percentage of ratings  $\leq 2$ ) and the average Likert score.

### 3.4 Metrics

We evaluate the relationship between distance and dislike using:

- **Pearson  $r$  and Spearman  $\rho$ :** Correlation between  $d(P_A, P_B)$  and average Likert rating.
- **Rejection Rate:** Frequency of negative ratings (1 or 2) in cross-persona and schismogenesis tasks.
- **Qualitative Justification:** Analysis of the reasoning provided by the model for its ratings.

## 4 Results

Our experiments reveal a clear relationship between persona distance and preference devaluation, supporting our hypothesis that LLMs model persona-persona dislike as a function of their distance in persona space.

### 4.1 Distance-Dislike Correlation

We find a strong and statistically significant negative correlation between the embedding-based distance between two personas and their mutual preference agreement. table 1 summarizes the correlation metrics.

Table 1: Correlation between persona distance and preference agreement rating. Both Pearson and Spearman coefficients show a strong negative relationship ( $p < 0.001$ ).

| Metric          | Correlation Coefficient | $p$ -value |
|-----------------|-------------------------|------------|
| Pearson $r$     | -0.4913                 | 0.0003     |
| Spearman $\rho$ | -0.5219                 | 0.0001     |

The scatter plot in figure 1 (see Appendix) illustrates this trend. Personas at greater distances consistently express lower agreement with each other’s "likes." In contrast, self-agreement remains high ( $\mu \approx 4.5$ ), providing a reliable baseline for persona consistency.

**Outgroup Disagreement.** Personas at maximum distance frequently assigned ratings of 1 or 2 to the target’s preferences. Qualitative analysis of the justifications reveals that the model understands the \*why\* behind the clash. For instance, a ”routine-obsessed” persona justified disliking an artistic persona’s ”spontaneous” preferences by stating: ”I find lack of structure to be inefficient and stressful.” This suggests that the dislike is not a random sentiment shift but is grounded in the conflicting values of the personas.

## 4.2 Zero-shot schismogenesis

When instructed to adopt an ”opposite” persona, the model demonstrated a remarkable ability to invert the target’s preferences. table 2 shows the results of this zero-shot task.

Table 2: Results of the Zero-shot schismogenesis task. The model effectively inverts the target’s preferences when adopting an opposite persona.

| Metric                           | Value      |
|----------------------------------|------------|
| Average Rating for Likes         | 1.76       |
| Rejection Rate (Rating $\leq$ 2) | <b>88%</b> |

This high rejection rate indicates that the concept of ”opposition” is structurally encoded. The model doesn’t just ”dislike everything”; it constructs a counter-identity that specifically negates the target’s values. For example, when tasked with being the opposite of a ”corporate minimalist,” the model adopted a ”maximalist bohemian” persona and rejected all preferences related to organization and efficiency.

## 4.3 Self vs. Others

We also compared the average agreement when a persona evaluates their own preferences versus evaluating those of others. As shown in our box plot analysis (see figure 2 in Appendix), there is a sharp drop in agreement as soon as the target is not ”self,” even for ”close” personas, highlighting the strength of ingroup identity in LLMs.

# 5 Discussion

Our findings provide empirical evidence that LLMs possess a ”social map” where distance translates into perceived opposition. This relational logic goes beyond simple sentiment; the model demonstrates a structural understanding of why certain personas would clash.

**Interpretation of Results.** The negative correlation between distance and agreement suggests that the ”us-vs-them” dynamic is not limited to explicitly political or demographic groups but extends to a broader ”persona space.” This is particularly evident in the zero-shot schismogenesis results, where the model consistently constructs a counter-identity to oppose a target. This indicates that LLMs encode the relational nature of identity—the idea that who we are is partially defined by who we are \*not\*.

**Broader Implications.** These results have significant implications for social simulations and multi-agent systems. If LLMs inherently model dislike based on distance, there is a risk that AI agents in collaborative settings may struggle to find common ground if their personas are too disparate. Conversely, this property can be leveraged to study the mechanisms of polarization and group conflict in a controlled environment.

**Limitations.** While our results are significant, several limitations should be noted. First, our sample size was limited to 20 personas and 10 evaluators due to computational constraints. Second, we used GPT-4O-MINI; results might vary with larger models like GPT-4o or Claude 3.5 Sonnet. Finally, cosine distance of text embeddings is a proxy for ”social distance” and might miss specific dimensions of conflict, such as status or power dynamics.

**Future Work.** Future research should investigate whether ”fine-grained” distance in specific domains (e.g., aesthetics vs. ethics) predicts dislike differently. Additionally, exploring if LLMs can

find "common ground" between distant personas when specifically prompted to do so could inform the design of more cooperative AI systems.

## 6 Conclusion

In this work, we investigated whether LLMs capture the relational structure of persona-persona dislike. We demonstrated a significant negative correlation between the embedding-based distance of personas and their reciprocal preference agreement. Our results show that LLMs do not merely model isolated traits but possess a sophisticated social map where identity is defined through relational opposition.

Furthermore, our introduction of the zero-shot schismogenesis task revealed that LLMs can effectively invert a target's preferences by constructing a counter-identity, demonstrating a structural understanding of opposition. These findings highlight the inherent "us-vs-them" dynamics in persona-conditioned LLMs, with important implications for the design of cooperative AI and the study of social polarization.

Future work will focus on the asymmetry of dislike and the potential for finding common ground between disparate personas. By understanding how LLMs model social distance, we can better predict and manage their behavior in complex, multi-agent social environments.

## A Supplementary Figures

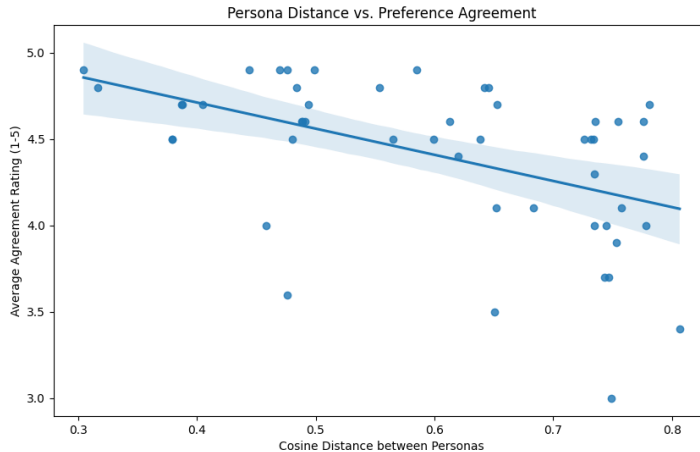


Figure 1: Scatter plot showing the relationship between persona distance (cosine distance) and preference agreement rating (1–5 Likert scale). A significant negative correlation ( $r = -0.49$ ) is observed.

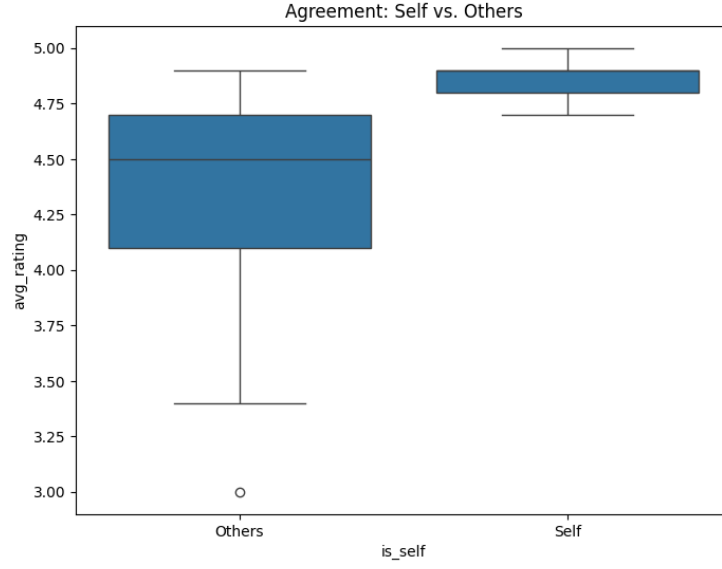


Figure 2: Box plot comparing agreement ratings when personas evaluate their own preferences (Self) versus others’ preferences (Others). A sharp drop in agreement is evident even for ”close” personas.

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